

Development and Characterization of a Robotic Hand and Forearm Using McKibben Muscles

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Abstract

This report presents the development and characterization of a pneumatic actuation approach for replacing the servo motors of an InMoov robotic hand with McKibben muscles (PAMs). The work includes in-house fabrication of McKibben actuators, experimental characterization using three dedicated rigs, integration of one PAM and one return spring per finger, forearm packaging for five independent finger units, and development of a five-channel proportional pressure regulation (PPR) circuit. The actuator was characterized through free contraction versus pressure, blocked force versus pressure, and static load-displacement stiffness tests. The results show repeatable pressure-dependent contraction, a nearly single-valued blocked-force response with low hysteresis, and pressure-tunable stiffness. These findings support the use of McKibben PAMs as compliant, lightweight actuators for safer and more adaptable robotic hand actuation.

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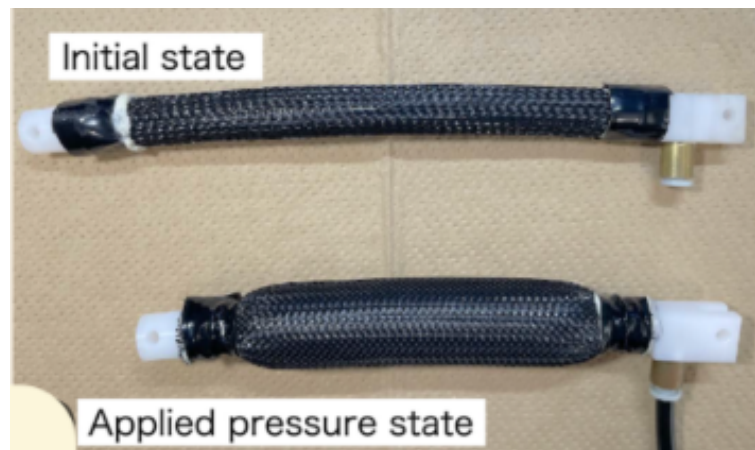
1 Introduction

The InMoov robotic hand is commonly actuated using servo motors. Although servo motors are compact and easy to command, they are rigid, relatively heavy at the hand/forearm scale, and provide limited inherent compliance. For assistive, wearable, and human-interactive robotic devices, compliance is important because the actuator should tolerate external resistance, accidental contact, and uncertain object geometry without producing unsafe or jerky motion.

McKibben pneumatic artificial muscles offer a soft actuation alternative. A McKibben muscle contracts axially when pressurized, generating tensile force in a manner similar to a biological muscle. This makes it suitable for tendon-driven finger flexion, especially when paired with a passive return element such as a spring. However, PAMs are nonlinear and may show hysteresis due to bladder elasticity, braid-bladder interaction, internal friction, and pneumatic effects. Therefore, experimental characterization is necessary before integrating the actuator into the hand.



(a) InMoov robotic hand.



(b) McKibben muscle in unpressurized and pressurized states.

Figure 1: Target robotic hand platform and the McKibben pneumatic artificial muscle used as the replacement actuator.

2 Project Objective and Scope

The objective of this work is to replace the servo-motor actuation of the InMoov robotic hand with McKibben muscles. The intended actuation architecture uses one McKibben muscle to flex each finger and one passive return spring to extend the finger after pressure release.

The work was divided into the following tasks:

1. Understand the operating principle of McKibben muscles and fabricate a functional PAM.
2. Build experimental rigs to characterize the fabricated PAM mechanically.
3. Measure free contraction, blocked force, and pressure-dependent static stiffness.

4. Design a per-finger PAM-spring actuation unit for the robotic hand.
5. Design the forearm arrangement for routing five independent PAM units.
6. Develop a five-channel PPR control circuit for independent pressure control of the five fingers.

3 McKibben Muscle Background

3.1 Operating Principle

A McKibben pneumatic artificial muscle consists of an inner elastomeric bladder surrounded by a braided sleeve and sealed at both ends by rigid end fittings. When compressed air is supplied to the bladder, the bladder expands radially. The surrounding braid resists this radial expansion and converts it into axial contraction. As a result, the actuator behaves as a tensile actuator: it pulls when pressurized and relaxes when depressurized.

Two important output measures are used throughout this report:

- **Free contraction:** the axial shortening of the actuator when it is pressurized without external load.
- **Blocked force:** the tensile force generated when the actuator is pressurized while its length is held fixed.

The free contraction ratio is calculated as

$$\epsilon = \frac{L_0 - L(P)}{L_0} \times 100, \quad (1)$$

where L_0 is the unpressurized actuator length and $L(P)$ is the length at pressure P .

3.2 Why McKibben Muscles Instead of Servo Motors?

The motivation for replacing servo motors with McKibben muscles is based on compliance, weight, and interaction safety. Servo motors provide direct position control but are relatively rigid and can transmit abrupt loads through the finger structure. In contrast, a McKibben muscle provides a compliant tensile actuation mechanism that can deform under external resistance.

The main advantages of the McKibben approach for this robotic hand are:

- high power-to-weight ratio;
- inherent compliance, which is useful for safe interaction and grasp adaptation;
- muscle-like tensile actuation suitable for tendon-driven finger flexion;
- simple construction using a bladder, braided sleeve, end fittings, and pneumatic tubing;
- low-cost and replaceable actuator components.

The main limitation is that the actuator response is nonlinear and pressure dependent. Therefore, characterization is required before reliable hand integration and control.

4 Actuator Fabrication

The PAM used in this work was fabricated in-house. The actuator consists of an inner latex bladder, a nylon braided sleeve, custom 3D-printed PLA end fittings, a pneumatic barbed fitting, and hose clips for sealing.



Figure 2: Fabrication components of the McKibben muscle, including the bladder tube, braided sleeve, pneumatic fitting, and hose clips.

Table 1: Fabricated PAM specifications.

Parameter	Value
Nominal unpressurized length	160 mm
Bladder material	Latex
Braid material	Nylon
Operating pressure range	0 kPa to 300 kPa
End fitting material	PLA, 3D printed

The bladder provides the sealed pressurized volume, while the braided sleeve defines the contraction behavior by restricting radial expansion. The end fittings and hose clips are critical because leaks at the ends directly affect both contraction and force output. The pneumatic port connects the actuator to the regulated pressure supply.

5 Experimental Characterization Methodology

Three independent characterization tests were performed to understand the behavior of the fabricated PAM. The tests were selected to capture the actuator response under free motion, fixed-length force generation, and externally loaded quasi-static conditions.

Table 2: Summary of characterization experiments.

Test	Measured quantity	Purpose
Displacement-pressure test	Free contraction ratio as a function of pressure	Determine how much actuator shortening is available for finger tendon pulling.
Blocked-force test	Isometric force as a function of pressure	Determine the maximum tensile force available when actuator length is constrained.
Static load-displacement test	Load-displacement slope at fixed pressures	Estimate pressure-dependent quasi-static stiffness and compliance.

5.1 Displacement-Pressure Characterization

5.1.1 Experimental Setup

The displacement-pressure test measures the unloaded axial contraction of the PAM as pressure is varied. A webcam was used with ArUco fiducial markers attached to the actuator ends. Python 3.12 and OpenCV were used for marker detection, frame-by-frame length measurement, and live plotting. Pixel measurements were converted to physical units using an in-frame calibration reference.

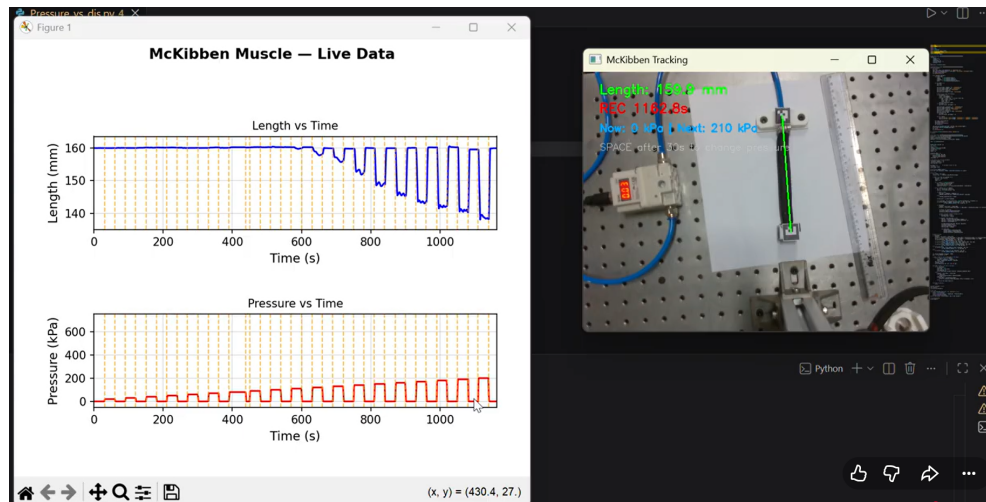


Figure 3: Optical tracking setup with live plots of muscle length versus time and pressure versus time.

Two displacement tests were performed. In the first test, the pressure was cycled through repeated steps such as 30 kPa $\rightarrow$ 0 kPa $\rightarrow$ 40 kPa $\rightarrow$ 0 kPa, and so on. This test checked whether the actuator returned to its baseline length after depressurization. In the second test, the pressure was swept upward and downward to evaluate displacement hysteresis between pressurization and depressurization branches.

5.1.2 Results

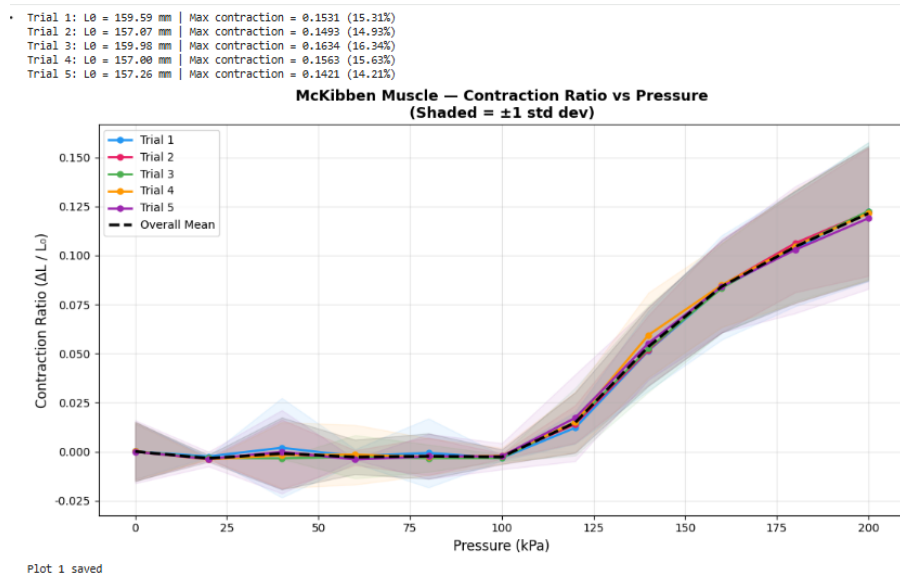


Figure 4: Contraction ratio versus pressure for five trials. The plot shows repeatability across trials and a pressure threshold before substantial contraction begins.

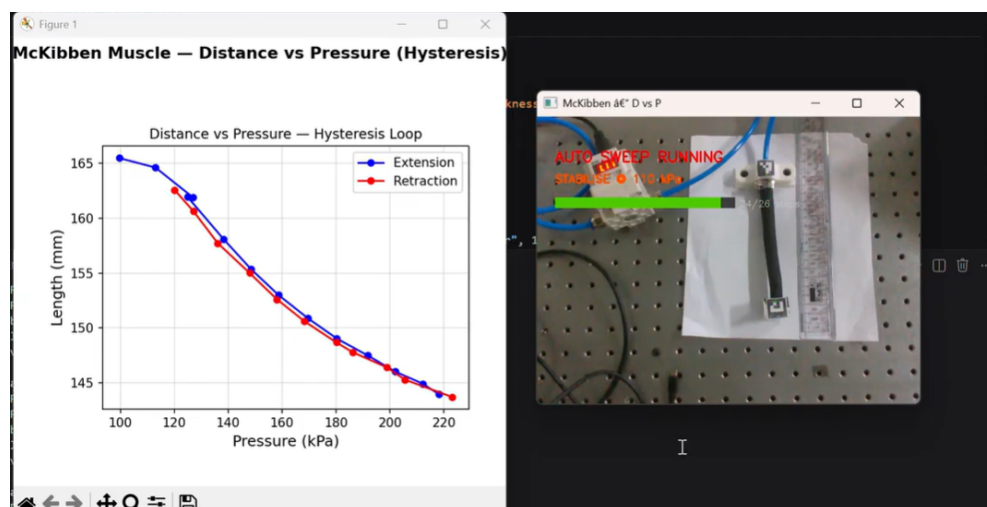


Figure 5: Live plotting of the length-pressure hysteresis loop during ascending and descending pressure sweep.

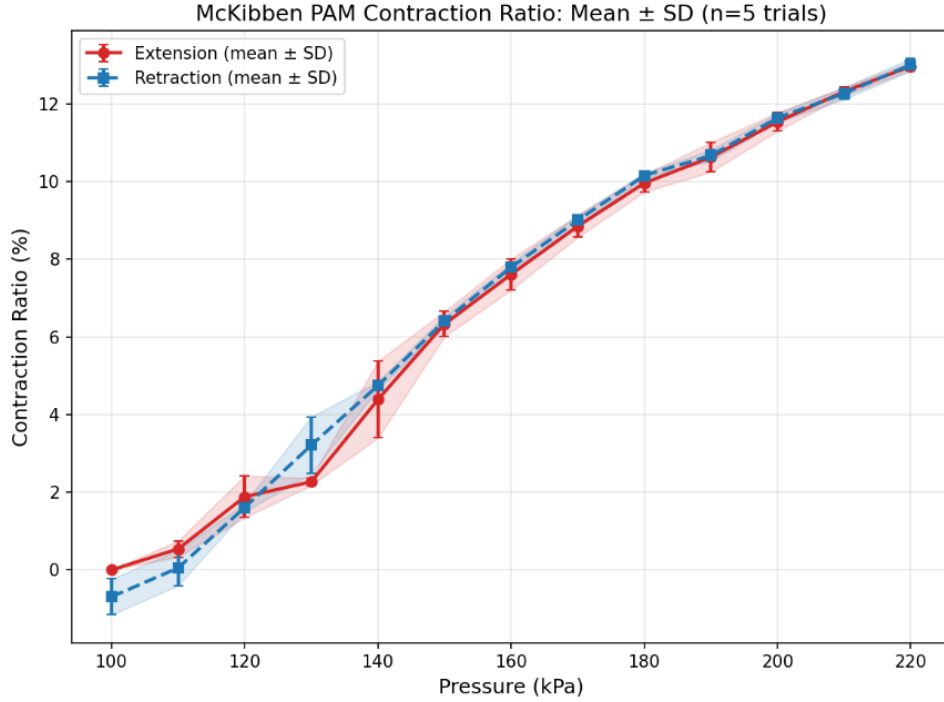


Figure 6: Mean contraction ratio versus pressure hysteresis over five trials.

Across five trials, the maximum free contraction was approximately 14.2% to 16.3%, with initial measured lengths of about 157 mm to 160 mm. The contraction ratio remained close to zero up to approximately 100 kPa; beyond this threshold, contraction increased monotonically with pressure. Between about 100 kPa to 220 kPa, the contraction-pressure trend was approximately linear and did not show a clear saturation plateau within the tested range.

The extension and retraction branches nearly overlapped, indicating low displacement hysteresis under the automated pressure sweep. The repeated low-pressure cycling also showed that the actuator returned to its baseline length without noticeable drift or creep over the tested duration.

5.1.3 Interpretation

The displacement-pressure experiment shows that the fabricated PAM provides repeatable contraction suitable for tendon pulling. The observed dead-zone up to approximately 100 kPa is important for control because small pressure commands below this threshold may not produce meaningful finger motion. The low hysteresis observed in the optical test suggests that the automated smooth pressure sweep and non-contact measurement reduced artifacts that may arise from manual pressure actuation or contact displacement sensors.

For hand integration, the key design implication is that a finger tendon pull of approximately 15 mm can be achieved from the actuator's free contraction range, but the required pressure increases once the actuator must also overcome return spring forces, tendon friction, and finger joint resistance.

5.2 Blocked-Force Characterization

5.2.1 Experimental Setup

The blocked-force test measures the force generated by the actuator when its length is held fixed. A horizontal rig was built using a FUTEK LCM300 load cell. The PAM was mounted between

a fixed anchor and the load cell so that pressurization produced tensile force without allowing axial shortening.

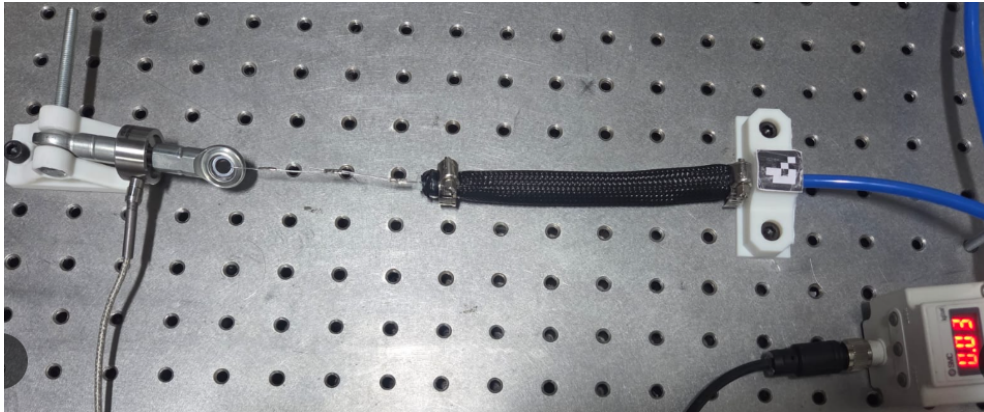


Figure 7: Horizontal blocked-force test rig using a FUTEK LCM300 load cell.

Pressure was ramped from 100 kPa to 220 kPa in steps while keeping the muscle length fixed. Force was logged at each step for both ascending and descending pressure branches. Crash-safe CSV logging and recovery from USB dropout were used during automated sweeps.

5.2.2 Results

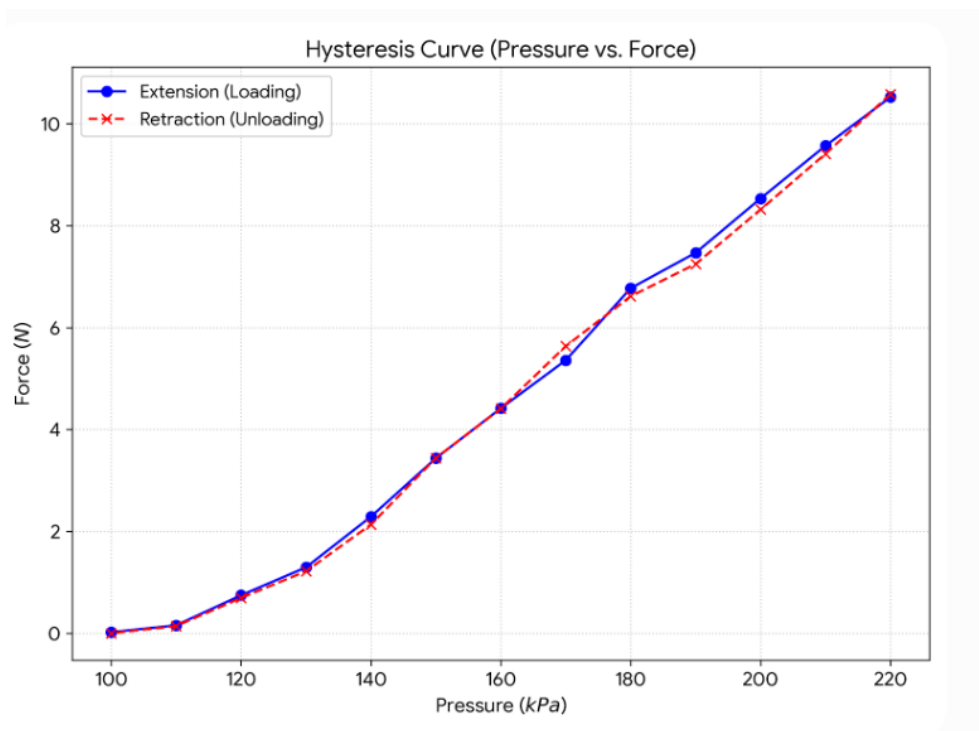


Figure 8: Blocked force versus supply pressure for ascending and descending pressure branches.

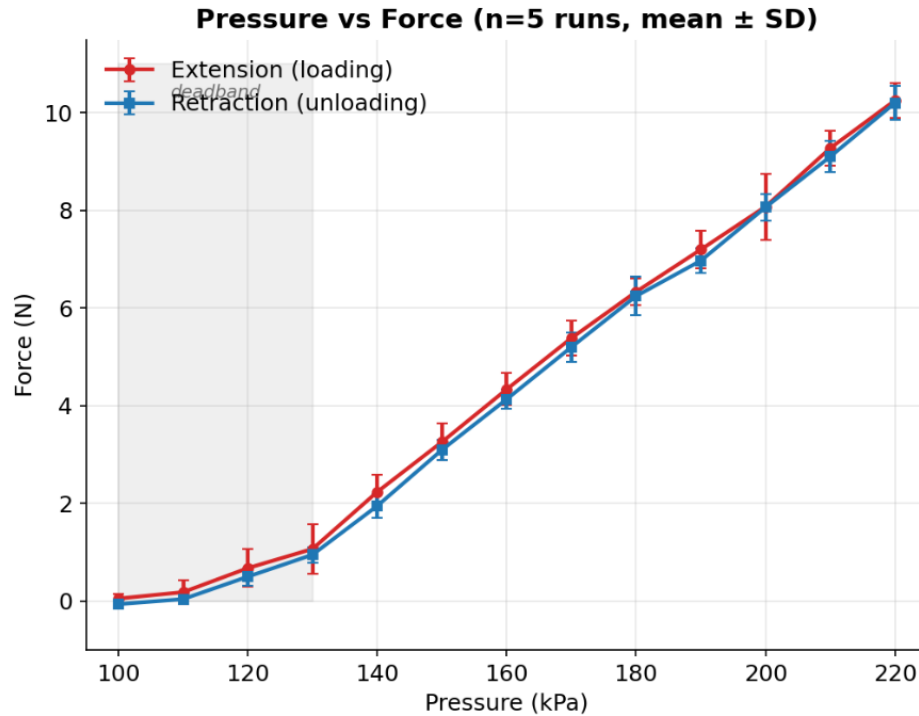


Figure 9: Blocked force versus supply pressure averaged over five trials. Error bars represent standard deviation.

The blocked-force response increased from approximately 0 N near 100 kPa to approximately 10.2 N at 220 kPa. The loading and unloading branches were nearly coincident, indicating minimal force hysteresis in the blocked condition. The average force-pressure slope in the working range was approximately 0.099 N/kPa, with a reported near-linear fit quality of $R^2 \approx 0.999$, although the plotted trend shows slight upward curvature at higher pressures.

5.2.3 Interpretation

The blocked-force data confirm that the actuator can generate meaningful tensile force over the operating pressure range. The small hysteresis is favorable for open-loop pressure control because the delivered force is approximately a single-valued function of pressure under the tested conditions. The convex trend is consistent with the nonlinear nature of McKibben muscle mechanics, where pressure, braid angle, and actuator strain jointly determine output force.

A larger bladder diameter would be expected to increase force output because pressure acts over a larger effective cross-sectional area. However, any change in diameter would also modify contraction behavior, stiffness, and packaging constraints inside the forearm.

5.3 Static Load-Displacement Stiffness Characterization

5.3.1 Experimental Setup

The static stiffness test characterizes actuator compliance at fixed pressure levels. The PAM was mounted vertically, and known weights were applied incrementally. The displacement was measured optically at each load step.



Figure 10: Vertical weight-stack rig for static load-displacement stiffness measurement.

At each fixed pressure setpoint, loads of 0 g, 250 g, 500 g, 750 g, and 1000 g were applied. The force-displacement relationship was used to estimate quasi-static stiffness as

$$k = \frac{\Delta F}{\Delta x}, \quad (2)$$

where ΔF is the applied force increment and Δx is the corresponding displacement change.

5.3.2 Results

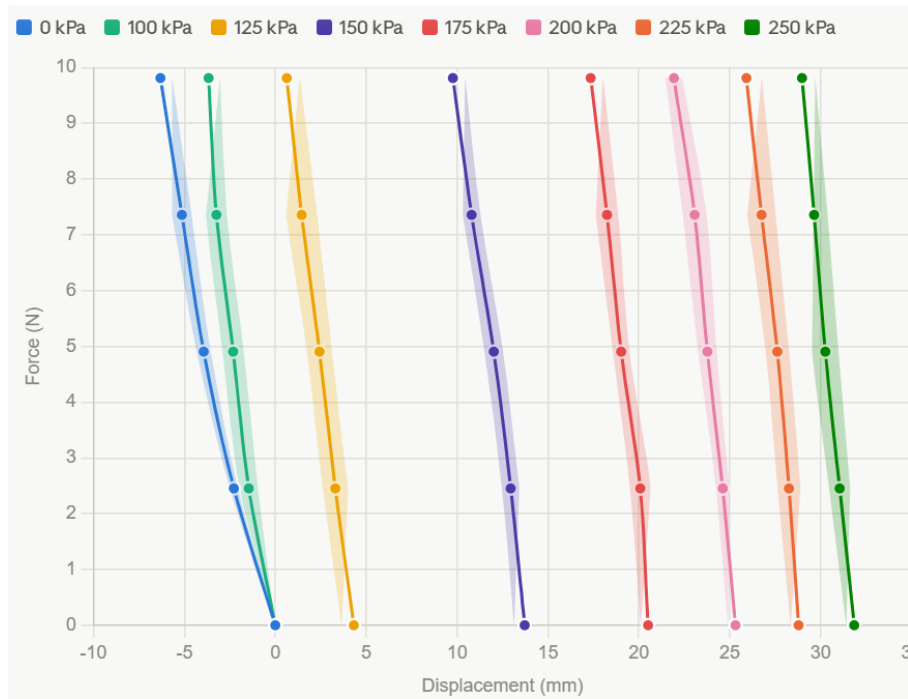


Figure 11: Load-displacement (contraction) response at varying supply pressures.

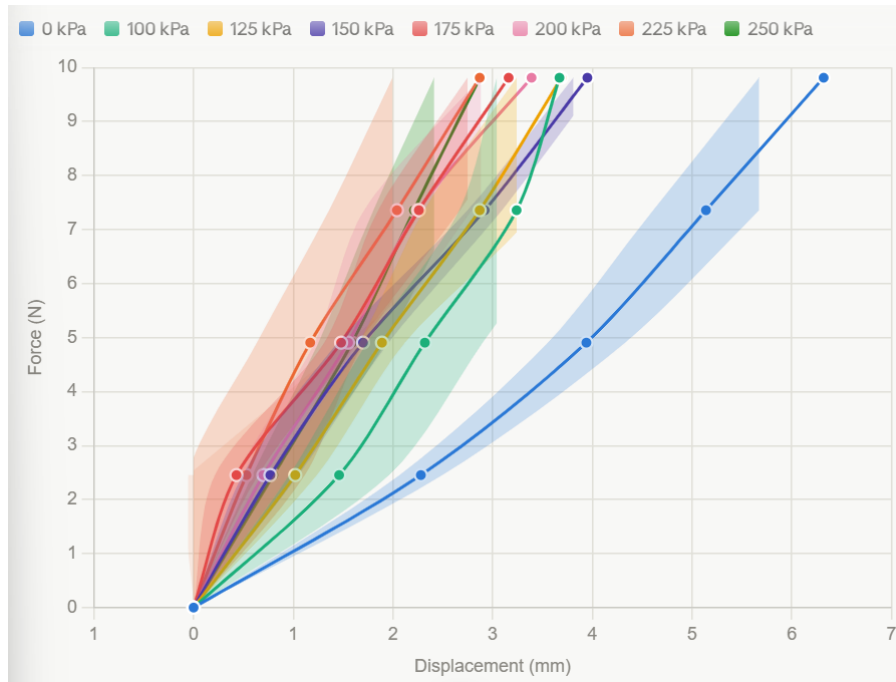


Figure 12: Stiffness trend at varying pressure. The slope of each force-displacement curve represents quasi-static stiffness.

At low or zero pressure, the actuator was highly compliant. Under an approximately 10 N load, displacement reached about 6 mm. At higher pressures, particularly around 200 kPa to 250 kPa, the same load produced only about 1 mm displacement. Therefore, the actuator stiffness increased markedly with pressure.

5.3.3 Interpretation

For each fixed pressure, the load-displacement response was approximately linear over the tested load range. This allows each pressure condition to be represented by an effective quasi-static stiffness. As pressure increased, the force-displacement curves became steeper, showing that the PAM becomes progressively less compliant. The largest stiffness change occurred when increasing pressure from 0 kPa into the lower pressurized range; at higher pressures the stiffness increase showed diminishing returns.

This pressure-tunable stiffness is a major advantage over a rigid servo-motor drive. In a robotic hand, the same actuator can be made more compliant for gentle interaction or stiffer for stronger grasping by adjusting the pressure setpoint.

6 Summary of Characterization Results

Table 3: Key experimental observations from the fabricated PAM.

Experiment	Reported result	Design implication
Free contraction	Maximum contraction of approximately 14.2 % to 16.3 %; threshold near 100 kPa.	Useful for tendon pulling, but low pressures may not produce motion.
Displacement hysteresis	Pressurization and depressurization branches nearly overlap.	Pressure can be used as a practical open-loop command variable in the tested range.
Blocked force	Force rises to approximately 10.2 N at 220 kPa; low hysteresis over five trials.	Actuator provides enough tensile force for experimental finger actuation, subject to spring and friction loads.
Static stiffness	Displacement under 10 N decreases from about 6 mm at low/zero pressure to about 1 mm at 200 kPa to 250 kPa.	Compliance can be tuned through pressure, which is valuable for safe grasping.

7 Finger and Forearm Design

7.1 Per-Finger PAM-Spring Mechanism

Each finger is designed to use one McKibben PAM and one return spring. The PAM provides active flexion when pressurized, while the spring provides passive extension when pressure is released. This arrangement is analogous to an agonist-passive antagonist pair.

- **Flexion actuation:** the McKibben muscle is connected to the finger tendon route so that axial contraction pulls the tendon and flexes the finger.
- **Extension return:** the spring stores energy during flexion and restores the finger to its initial position after depressurization.
- **Independent control:** each finger has its own PAM and pressure channel, allowing different pressure setpoints for different fingers.

The design avoids placing servo motors at each finger and shifts actuation to a compliant pneumatic architecture. The finger force and motion depend not only on the PAM contraction but also on tendon routing, joint friction, spring preload, and finger geometry.

7.2 Forearm Packaging

The forearm housing accommodates five independent finger units: one for the thumb and one for each of the four fingers. Each unit includes a pneumatic line, a PAM, tendon routing, and a return spring. The forearm also provides routing space so that tubes and tendons do not kink or interfere with moving joints.



Figure 13: Forearm assembly housing five PAM-spring actuation units.

Complete finger actuation requires approximately 15 mm of tendon/string pull. Based on the free-contraction characterization, this displacement is achievable around 200 kPa in the unloaded actuator. In the integrated hand, however, the PAM must also overcome the return spring, tendon friction, and joint resistance. Therefore, the required operating pressure in the assembled mechanism is higher, typically around 300 kPa. The required pressure is not identical for all fingers because each finger can have different spring preload, tendon friction, and geometry. Consequently, each pressure channel must be calibrated independently.

8 Five-Channel PPR Control Circuit

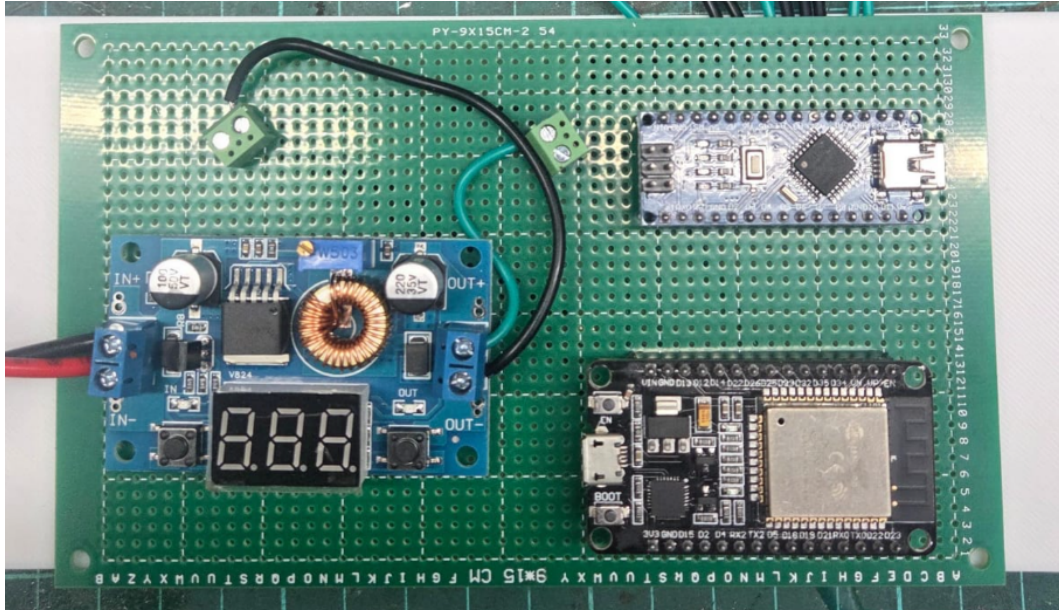
To actuate all five fingers independently, a five-channel proportional pressure regulation circuit was built. The system uses one regulator channel per finger, allowing individual pressure commands for the thumb and four fingers.

8.1 Circuit Architecture

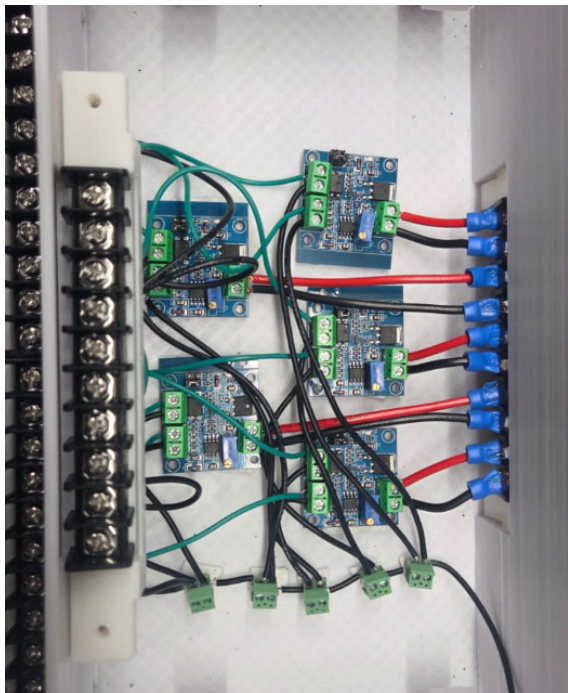
The circuit is built around an ESP32, an Arduino Nano, five PWM-to-analog modules, and five SMC ITV1050 proportional pressure regulators. A buck converter provides the required supply levels for the control electronics and pressure regulation hardware.

- **ESP32:** provides WiFi communication and an HTTP server interface for the PC-based control GUI. It communicates with the Arduino Nano over UART2 using GPIO16 and GPIO17 at 9600 baud.
- **Arduino Nano:** generates five independent PWM commands and reads five analog pressure feedback signals.

- **PWM-to-analog modules:** convert PWM signals into 0–8 V analog control voltages for the ITV1050 regulators.
- **SMC ITV1050 regulators:** produce regulated pneumatic pressure for each PAM and provide analog feedback voltage.
- **Feedback output:** the regulator monitor signal is 1–5 V corresponding to 0–900 kPa, which is read by the Nano.
- **Level shifting:** resistor-based level shifting protects the ESP32 UART pins from 5 V Nano logic.



(a) Main five-channel pressure regulation hardware.



(b) Control electronics and wiring.



(c) PPR regulator assembly.

Figure 14: Five-channel PPR control circuit for independent pneumatic actuation of the five fingers.

8.2 Control Operation

The system operation can be summarized as follows:

1. The PC-side Python GUI sends a desired pressure setpoint for a channel to the ESP32 using an HTTP `POST /set` request.
2. The ESP32 forwards the command to the Arduino Nano over UART as a `SET:<value>` command.
3. The Nano checks whether the requested pressure lies within the allowed 0–900 kPa range and returns an acknowledgement such as `ACK:<value>`.
4. The Nano converts the pressure command into a PWM duty cycle between 0 and 255.
5. The PWM-to-analog module converts the duty cycle into a 0–8 V analog control signal for the ITV1050 regulator.
6. The regulator outputs the corresponding pneumatic pressure to the assigned PAM.
7. The regulator feedback voltage is read by the Nano and converted back into a pressure value.
8. Every 500 ms, the Nano sends the state information to the ESP32 using the serial format `DATA:setpoint:actual:duty`.
9. The ESP32 exposes live feedback through HTTP `GET /status`, allowing the GUI to monitor and log commanded and measured pressure.

This architecture is appropriate for the PAM-spring hand because each finger can require a different operating pressure. Independent pressure channels allow finger-specific calibration and compensation for differences in spring stiffness, tendon routing, and friction.

9 Discussion

The experimental characterization confirms that the fabricated McKibben muscle is suitable for use as a compliant tendon actuator in the robotic hand. The displacement-pressure results show that the actuator can provide the required order of contraction for finger tendon pulling. The blocked-force results show that the muscle can generate measurable tensile force with low force hysteresis under the tested conditions. The static stiffness results demonstrate that the actuator stiffness can be adjusted by pressure.

The most important design observation is that free-actuator pressure requirements do not directly equal integrated-hand pressure requirements. In the isolated test, pressure mainly produces contraction. In the hand, pressure must produce contraction while also overcoming spring preload, tendon friction, joint friction, and load from contact with objects. Therefore, a finger may require around 300 kPa during assembly-level operation even if the free actuator reaches the target contraction near 200 kPa.

The low hysteresis observed in the reported plots is useful for control, but it should not be assumed to remain unchanged under all conditions. Integrated finger tests may show additional hysteresis due to tendon friction, joint backlash, spring effects, and tube dynamics. Thus, pressure control alone is a good starting point, but future versions should incorporate finger position or force feedback for robust grasping.

10 Limitations

The following limitations should be considered when interpreting the report:

- The characterization was primarily performed on the isolated PAM rather than the fully integrated hand.
- Finger-level effects such as tendon routing friction, joint friction, and spring preload were not separately quantified.
- Dynamic response, bandwidth, and pressure rise/fall time were not characterized in detail.
- Long-term fatigue and durability of the bladder, braid, fittings, and seals were not tested.
- The force and stiffness results apply to the fabricated actuator geometry and materials used in this work; changes in bladder diameter, braid angle, or end fitting design may alter the response.

11 Future Work

The next stage of this work should focus on integrated hand-level testing and control. Recommended future work includes:

- **Whole-hand characterization:** measure grip force, grasp repeatability, object holding stability, and finger-to-finger variation in the assembled hand.
- **Per-finger pressure calibration:** determine the minimum and operating pressure required for each finger to reach full flexion.
- **Closed-loop position control:** add finger position sensing to compensate for nonlinear contraction and tendon friction.
- **Closed-loop force control:** add fingertip or tendon force sensing to regulate grasping force and prevent excessive contact force.
- **Durability testing:** perform cyclic pressurization tests to quantify actuator fatigue, leakage, creep, and repeatability over time.
- **Vision-based actuation logic:** extend the existing OpenCV tracking workflow toward object detection and grasp planning for autonomous or semi-autonomous actuation.

12 Conclusion

This report presented the fabrication, characterization, and robotic-hand integration plan for McKibben pneumatic artificial muscles as replacements for servo motors in an InMoov robotic hand. The fabricated PAM was tested using optical displacement tracking, blocked-force measurement, and static load-displacement stiffness characterization. The actuator showed repeatable contraction, low hysteresis in the tested conditions, blocked force up to approximately 10.2N at 220 kPa, and pressure-tunable stiffness. A PAM-spring finger actuation concept and five-channel PPR circuit were developed to enable independent actuation of all five fingers. Overall, the results support McKibben muscles as a feasible compliant actuation approach for the robotic hand, with the next step being full hand-level calibration and closed-loop control.

Individual Contributions

Prachi Jindal led the experimental characterization, data analysis and mechanical development of actuation system.

- Development of the optical displacement-tracking setup using ArUco markers and OpenCV for the displacement-pressure characterization test.
- Design and execution of the pressure cycling and sweep test protocols (Section 5.1).
- Design and assembly of the blocked-force test rig (FUTEK LCM300 load cell setup) and the static load-displacement stiffness rig (weight-stack setup).
- Data processing, plotting, and statistical analysis (mean $\pm$ SD) for all three characterization experiments across five trials.
- Interpretation of hysteresis, contraction, and stiffness trends, and preparation of Figures 4–12.
- Report writing for Sections 5, 6, 9, and 10 (Experimental Methodology, Summary of Results, Discussion, Limitations).

Ayushi Meena led the mechanical and electrical development of the actuation system.

- Fabrication of the McKibben PAM actuators (bladder, braided sleeve, end fittings, sealing).
- Design of the five-channel PPR control circuit, including ESP32–Arduino Nano communication, PWM-to-analog interfacing, and SMC ITV1050 regulator integration.
- Firmware development for pressure command handling, PWM duty-cycle conversion, and feedback logging.
- Integration of the PAM-spring actuation mechanism into the finger and forearm assembly.
- Report writing for Sections 4, 7, and 8 (Actuator Fabrication, Finger/Forearm Design, PPR Control Circuit).

Both authors jointly contributed to the overall project planning, review of results, and finalization of the report.

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